

Class NOTES

Gourab Biswas

November 19, 2024

Class 1: GCD and Primes

Euclid's Algorithm

The Euclidean Algorithm calculates the greatest common divisor (gcd) of two integers m and n :

1. Divide m by n to get the remainder r : $m = nq + r$, where $0 \leq r < n$.
2. Replace m with n and n with r .
3. Repeat until $r = 0$. The last non-zero n is the $\gcd(m, n)$.

Example: Find $\gcd(252, 105)$:

$$252 = 105 \cdot 2 + 42, \quad 105 = 42 \cdot 2 + 21, \quad 42 = 21 \cdot 2 + 0.$$

Thus, $\gcd(252, 105) = 21$.

Extended Euclidean Algorithm

The Extended Euclidean Algorithm expresses the gcd of two integers m and n as a linear combination:

$$\gcd(m, n) = am + bn,$$

where $a, b \in \mathbb{Z}$.

Example: For $m = 252$ and $n = 105$:

$$252 = 105 \cdot 2 + 42,$$

$$105 = 42 \cdot 2 + 21,$$

$$42 = 21 \cdot 2 + 0.$$

Back-substitution gives:

$$21 = 105 - 42 \cdot 2, \quad 42 = 252 - 105 \cdot 2.$$

Substituting for 42:

$$21 = 105 - (252 - 105 \cdot 2) \cdot 2 = 105 \cdot 5 - 252 \cdot 2.$$

Thus, $\gcd(252, 105) = 21 = (-2) \cdot 252 + 5 \cdot 105$.

Fundamental Theorem of Arithmetic

Every positive integer $N > 1$ can be expressed as a product of primes:

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k},$$

where $p_1, p_2, \dots, p_k$ are distinct primes, and $a_1, a_2, \dots, a_k$ are positive integers. This factorization is unique up to the order of the primes.

Example: $120 = 2^3 \cdot 3^1 \cdot 5^1$.

Proof

Existence of Prime Factorization

1. **Base Case:** If N is prime, it is already a product of a single prime.
2. **Inductive Step:** Assume every positive integer $M < N$ can be expressed as a product of primes.
 - If N is not prime, it can be written as $N = ab$, where $1 < a, b < N$.
 - By the induction hypothesis, both a and b can be expressed as products of primes:

$$a = p_1^{b_1} p_2^{b_2} \cdots p_m^{b_m}, \quad b = q_1^{c_1} q_2^{c_2} \cdots q_n^{c_n}.$$

- Therefore, $N = ab$ can also be expressed as a product of primes by combining the prime factorizations of a and b .

Uniqueness of Prime Factorization

1. Suppose N has two distinct factorizations:

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} = q_1^{b_1} q_2^{b_2} \cdots q_l^{b_l},$$

where $p_1, p_2, \dots, p_k$ and $q_1, q_2, \dots, q_l$ are primes, and $p_i \neq q_j$ for some i, j

2. Let p_1 be the smallest prime in the factorization of N . Since p_1 divides N , it must divide one of the primes $q_1, q_2, \dots, q_l$.
3. But primes are indivisible except by themselves, so p_1 must equal one of q_j .
4. Divide N by p_1 , and repeat the argument for the remaining product. This process continues until all prime factors are exhausted.

Conclusion

Every positive integer N can be expressed as a product of primes in a unique way, up to the order of the primes.

Proof of Uniqueness using Bézout's Lemma

Step 1: Assume Two Factorizations

Suppose N has two factorizations:

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} = q_1^{b_1} q_2^{b_2} \cdots q_l^{b_l},$$

where $p_1 \leq p_2 \leq \cdots \leq p_k$ and $q_1 \leq q_2 \leq \cdots \leq q_l$.

Step 2: Use Bézout's Lemma

Since $p_1 \neq q_1$, $\gcd(p_1, q_1) = 1$. By Bézout's Lemma, there exist integers a, b such that:

$$ap_1 + bq_1 = 1.$$

Step 3: Multiply by $p_2 p_3 \cdots p_k$

Multiply both sides by $p_2 p_3 \cdots p_k$:

$$aN + bq_1(p_2 p_3 \cdots p_k) = (p_2 p_3 \cdots p_k).$$

- Left-hand side: aN and $bq_1(p_2 p_3 \cdots p_k)$ are divisible by q_1 .
- Right-hand side: $(p_2 p_3 \cdots p_k)$ is not divisible by q_1 , unless $q_1 = p_2, p_3, \dots, p_k$.

Thus, $p_1 \leq q_1$.

Step 4: Multiply by $q_2 q_3 \cdots q_l$

Similarly, multiplying $ap_1 + bq_1 = 1$ by $q_2 q_3 \cdots q_l$ gives:

$$aN + bq_1(q_2 q_3 \cdots q_l) = (q_2 q_3 \cdots q_l).$$

- Left-hand side: aN and $bq_1(q_2 q_3 \cdots q_l)$ are divisible by p_1 .
- Right-hand side: $(q_2 q_3 \cdots q_l)$ is not divisible by p_1 , unless $p_1 = q_2, q_3, \dots, q_l$.

Thus, $q_1 \leq p_1$.

Step 5: Combine Results

From $p_1 \leq q_1$ and $q_1 \leq p_1$, conclude $p_1 = q_1$.

Conclusion

By recursion, all primes and exponents match, proving uniqueness.

Proof of Infinite Primes

We prove that there are infinitely many primes by contradiction:

1. Assume a finite set of primes: $p_1, p_2, \dots, p_k$.
2. Consider the number $P = p_1 \cdot p_2 \cdots p_k + 1$.
3. P is not divisible by any p_i , as dividing P by p_i leaves a remainder of 1.

Thus, P is either prime or has a prime factor not in the set. This contradicts the assumption.

Upper Bound of $\epsilon_p(n!)$

The largest power of a prime p dividing $n!$ is:

$$\epsilon_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Example: Find $\epsilon_2(10!)$:

$$\epsilon_2(10!) = \left\lfloor \frac{10}{2} \right\rfloor + \left\lfloor \frac{10}{4} \right\rfloor + \left\lfloor \frac{10}{8} \right\rfloor = 5 + 2 + 1 = 8.$$

Exercise 1: Prove $\gcd(km, kn) = k \cdot \gcd(m, n)$

Let $\gcd(m, n) = d$, so $m = da$ and $n = db$ with $\gcd(a, b) = 1$. For km and kn :

$$km = k \cdot da, \quad kn = k \cdot db.$$

The common divisor of km and kn is $k \cdot d$, so:

$$\gcd(km, kn) = k \cdot \gcd(m, n).$$

Exercise 2: Express $\epsilon_p(n!)$ in Terms of $v_p(n)$

The power of p dividing $n!$ is:

$$\epsilon_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Exercise 3: Prove $\gcd(12n + 1, 30n + 2) = 1$

We aim to prove that $\gcd(12n + 1, 30n + 2) = 1$ for any integer n .

Let:

$$a = 12n + 1 \quad \text{and} \quad b = 30n + 2.$$

We will compute $\gcd(a, b)$ using the Euclidean algorithm.

Step 1: Compute $b - 2a$

$$b - 2a = (30n + 2) - 2(12n + 1) = 30n + 2 - 24n - 2 = 6n.$$

Thus:

$$\gcd(a, b) = \gcd(12n + 1, 6n).$$

Step 2: Compute $a - 2(6n)$

$$a - 2(6n) = (12n + 1) - 2(6n) = 12n + 1 - 12n = 1.$$

Thus:

$$\gcd(12n + 1, 6n) = \gcd(6n, 1).$$

Step 3: Compute $\gcd(6n, 1)$ The greatest common divisor of any number and 1 is 1. Hence:

$$\gcd(6n, 1) = 1.$$

Conclusion Using the Euclidean algorithm, we have shown that:

$$\gcd(12n + 1, 30n + 2) = 1.$$

This proves that $12n + 1$ and $30n + 2$ are coprime for any integer n .

Class 2: Stern Brocot and Modular Arithmetic

Stern-Brocot Tree

The Stern-Brocot Tree is a binary tree of all positive fractions $\frac{p}{q}$ in their reduced form, where $p, q \in \mathbb{N}$, constructed as follows:

- The tree starts with two "anchor" fractions: $\frac{0}{1}$ on the left and $\frac{1}{0}$ on the right.
- Each subsequent node (called the mediant) is constructed by:

$$\text{mediant} = \frac{p_1 + p_2}{q_1 + q_2},$$

where $\frac{p_1}{q_1}$ and $\frac{p_2}{q_2}$ are adjacent fractions.

Why Each Mediant Fraction is in Lowest Terms

A mediant $\frac{p_1+p_2}{q_1+q_2}$ is in lowest terms because:

1. $p_1q_2 - p_2q_1 = \pm 1$ for adjacent fractions (property of the tree), implying the numerator and denominator are relatively prime.

Why Do All Possible Fractions Occur Exactly Once?

Every fraction $\frac{p}{q}$ appears exactly once because:

- The mediant operation ensures all intermediate fractions between two adjacent fractions are generated without repetition.
- This is guaranteed by the Farey sequence property and recursive construction.

Can Any Fraction Be Missing from This Tree?

No fraction in lowest terms is missing because:

- The tree exhaustively generates every fraction between $\frac{0}{1}$ and $\frac{1}{0}$.

Introduction to Modular Arithmetic

Modular arithmetic is a system of arithmetic for integers, where numbers "wrap around" upon reaching a certain value (the modulus). For $a, b \in \mathbb{Z}$ and modulus $m > 0$:

$$a \equiv b \pmod{m} \quad \text{if } m \mid (a - b).$$

Basic Operations

For integers $a, b, c \in \mathbb{Z}$:

- Addition: $(a + b) \pmod{m} = ((a \pmod{m}) + (b \pmod{m})) \pmod{m}$.
- Subtraction: $(a - b) \pmod{m} = ((a \pmod{m}) - (b \pmod{m})) \pmod{m}$.
- Multiplication: $(a \cdot b) \pmod{m} = ((a \pmod{m}) \cdot (b \pmod{m})) \pmod{m}$.

Why Division Does Not Work?

Division in modular arithmetic is not always defined because it requires the divisor to have a multiplicative inverse modulo m . For $b \cdot x \equiv 1 \pmod{m}$, the inverse exists if and only if $\gcd(b, m) = 1$.

Special Cases of Division

If $\gcd(b, m) = d > 1$, division modulo m is only possible when $a \equiv 0 \pmod{d}$.

Exercises

Exercise 1: Divisibility by 3 via Sum of Digits

A number in decimal notation is divisible by 3 if and only if the sum of its digits is divisible by 3.

Proof: Let the number N be represented in base 10 as:

$$N = a_k \cdot 10^k + a_{k-1} \cdot 10^{k-1} + \cdots + a_0,$$

where a_i are the digits of N .

Modulo 3, we note that:

$$10 \equiv 1 \pmod{3}, \quad 10^2 \equiv 1 \pmod{3}, \quad \text{and so on.}$$

Thus:

$$N \equiv a_k + a_{k-1} + \cdots + a_0 \pmod{3}.$$

This means N is divisible by 3 if and only if the sum of its digits $\sum a_i$ is divisible by 3.

Exercise 2: $p \equiv 3 \pmod{4}$ and $(n^2 + 1)$

Prove that if $p \equiv 3 \pmod{4}$, there is no integer n such that $p \mid (n^2 + 1)$.

Proof: Assume $p \mid (n^2 + 1)$. Then:

$$n^2 + 1 \equiv 0 \pmod{p} \implies n^2 \equiv -1 \pmod{p}.$$

This implies -1 is a quadratic residue modulo p . However:

- If $p \equiv 3 \pmod{4}$, it is known that -1 is not a quadratic residue modulo p (by properties of the Legendre symbol).

Thus, no such n exists.

Exercise 3: $n^j \equiv 1 \pmod{m}$ and $n^i \equiv 1 \pmod{m}$

If $n^j \equiv 1 \pmod{m}$ and $n^i \equiv 1 \pmod{m}$, then $n^{\gcd(i,j)} \equiv 1 \pmod{m}$.

Proof: Let $d = \gcd(i, j)$. Then:

$$d = a \cdot i + b \cdot j \quad \text{for integers } a, b.$$

Raising n to the power d :

$$n^d = n^{a \cdot i + b \cdot j} = (n^i)^a \cdot (n^j)^b.$$

Since $n^i \equiv 1 \pmod{m}$ and $n^j \equiv 1 \pmod{m}$, we have:

$$n^d \equiv 1 \pmod{m}.$$

Exercise 4: Divisibility of $n(n+1)(2n+1)$ by 6

Prove that $n(n+1)(2n+1)$ is always divisible by 6 for any integer n .

Proof: The expression $n(n+1)(2n+1)$ is the product of three consecutive integers. Consider:

- n and $n+1$ are consecutive integers, so one of them is divisible by 2.
- Among n , $n+1$, and $2n+1$, one is divisible by 3 (by the Pigeonhole Principle modulo 3).

Thus, the product $n(n+1)(2n+1)$ is divisible by both 2 and 3, and hence divisible by 6.

Class 3 - Residue Number System and Fermat's Little Theorem

Residue Number System

The Residue Number System (RNS) represents integers using their remainders when divided by a set of pairwise coprime moduli.

Properties of Residue Number System

Let the moduli be $m_1, m_2, \dots, m_k$, where $\gcd(m_i, m_j) = 1$ for $i \neq j$. Any integer x can be uniquely represented as:

$$(x \bmod m_1, x \bmod m_2, \dots, x \bmod m_k).$$

The operations (addition, subtraction, multiplication) are performed independently in each modulus.

Solutions to $x^2 \equiv 1 \pmod{m}$

We solve $x^2 \equiv 1 \pmod{m}$, where m can be factored into prime powers:

$$m = p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k}.$$

Case 1: Prime Modulus $m = p$

For a prime p , $x^2 \equiv 1 \pmod{p}$ has two solutions:

$$x \equiv 1 \pmod{p} \quad \text{and} \quad x \equiv -1 \pmod{p}.$$

Case 2: Composite Modulus m

If m is composite, the number of solutions is determined by the Chinese Remainder Theorem (CRT):

- For each prime power $p_i^{e_i}$, there are two solutions ($x \equiv 1 \pmod{p_i^{e_i}}$ and $x \equiv -1 \pmod{p_i^{e_i}}$).
- The total number of solutions is 2^k , where k is the number of distinct prime factors of m .

Fermat's Little Theorem

Statement: If p is a prime number and a is an integer such that $p \nmid a$, then:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof of Fermat's Little Theorem

Let the residue classes modulo p be $[1], [2], \dots, [p-1]$. Multiplying all residue classes by a , we get:

$$[a \cdot 1], [a \cdot 2], \dots, [a \cdot (p-1)].$$

Since $p \nmid a$, a has a multiplicative inverse modulo p , so these products are simply a rearrangement of $[1], [2], \dots, [p-1]$. Thus:

$$[a \cdot 1] \cdot [a \cdot 2] \cdots [a \cdot (p-1)] \equiv [1] \cdot [2] \cdots [p-1] \pmod{p}.$$

Canceling the common terms $[1] \cdot [2] \cdots [p-1]$, we find:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Exercises

Exercise 1: Compute $(2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60}) \pmod{7}$

We compute each term modulo 7 separately, using Fermat's Little Theorem ($a^6 \equiv 1 \pmod{7}$).

- $2^{20} \pmod{7}$:

$$2^{20} = (2^6)^3 \cdot 2^2 \equiv 1^3 \cdot 4 \equiv 4 \pmod{7}.$$

- $3^{30} \pmod{7}$:

$$3^{30} = (3^6)^5 \equiv 1^5 \equiv 1 \pmod{7}.$$

- $4^{40} \pmod{7}$:

$$4^{40} = (4^6)^6 \cdot 4^4 \equiv 1^6 \cdot (4^2)^2 \equiv 1 \cdot 2^2 \equiv 4 \pmod{7}.$$

- $5^{50} \pmod{7}$:

$$5^{50} = (5^6)^8 \cdot 5^2 \equiv 1^8 \cdot 4 \equiv 4 \pmod{7}.$$

- $6^{60} \pmod{7}$:

$$6^{60} = (6^6)^{10} \equiv 1^{10} \equiv 1 \pmod{7}.$$

Adding these results:

$$2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60} \equiv 4 + 1 + 4 + 4 + 1 \equiv 14 \equiv 0 \pmod{7}.$$

Final Answer: $0 \pmod{7}$.

Exercise 2: Find p such that p and $8p^2 + 1$ are primes

Let p and $8p^2 + 1$ be primes. We analyze the given equation:

$$q = 8p^2 + 1 \pmod{3}$$

For any p :

$$p^2 \equiv 0 \text{ or } 1 \pmod{3}.$$

Case 1: $p^2 \equiv 0 \pmod{3}$

If $p^2 \equiv 0 \pmod{3}$, then $p \equiv 0 \pmod{3}$, meaning $p = 3$ (since p is prime). Substituting $p = 3$ into the equation for q :

$$q = 8(3^2) + 1 = 8(9) + 1 = 72 + 1 = 73.$$

Here, $q = 73$, which is a prime number. Thus, $p = 3$ satisfies the condition.

Case 2: $p^2 \equiv 1 \pmod{3}$

If $p^2 \equiv 1 \pmod{3}$, substitute this into the expression for q :

$$q = 8p^2 + 1 \equiv 8(1) + 1 \equiv 9 \equiv 0 \pmod{3}.$$

This implies q is divisible by 3. For q to be prime, it must be $q = 3$. If $q = 3$, then:

$$8p^2 + 1 = 3 \implies 8p^2 = 2 \implies p^2 = \frac{1}{4}.$$

However, $p^2 = \frac{1}{4}$ is not possible because p^2 must be an integer.

Thus, the assumption $p^2 \equiv 1 \pmod{3}$ leads to a contradiction unless $p = 3$.

Conclusion

From the above analysis:

- If $p^2 \equiv 0 \pmod{3}$, then $p = 3$ satisfies the conditions, and $q = 73$ is prime.
- If $p^2 \equiv 1 \pmod{3}$, q is divisible by 3 and cannot be prime.

Thus, the only solution is $p = 3$.

Final Answer: $p = 3$.

Class 4: Euler's Totient Function

Introduction to Euler's Totient Function

Euler's Totient Function, denoted by $\phi(n)$, is defined as the number of integers k such that $1 \leq k \leq n$ and $\gcd(k, n) = 1$. Mathematically:

$$\phi(n) = |\{k \in \mathbb{Z}^+ \mid 1 \leq k \leq n, \gcd(k, n) = 1\}|.$$

Examples

- $\phi(1) = 1$, because only 1 is relatively prime to 1.
- $\phi(2) = 1$, because only 1 is relatively prime to 2.
- $\phi(6) = 2$, because 1 and 5 are the only integers less than 6 that are relatively prime to 6.
- $\phi(12) = 4$, because 1, 5, 7, 11 are relatively prime to 12.

Key Theories and Proofs

1. $\phi(p) = p - 1$ for a Prime p

Statement: If p is a prime number, then:

$$\phi(p) = p - 1.$$

Proof: 1. For a prime p , all integers $1, 2, \dots, p - 1$ are relatively prime to p , since $\gcd(k, p) = 1$ for all $k < p$. 2. Therefore, there are $p - 1$ integers that satisfy the condition $\gcd(k, p) = 1$. 3. Hence:

$$\phi(p) = p - 1.$$

2. $\phi(p^k) = p^k - p^{k-1}$ for Prime Powers

Statement: For a prime power p^k , the number of integers less than or equal to p^k that are relatively prime to p^k is:

$$\phi(p^k) = p^k - p^{k-1}.$$

Proof: 1. The total number of integers less than or equal to p^k is p^k . 2. Out of these, the integers divisible by p are $p, 2p, 3p, \dots, p^{k-1}p$. 3. There are exactly p^{k-1} multiples of p in this range. 4. Subtracting the multiples of p , the remaining integers are relatively prime to p^k :

$$\phi(p^k) = p^k - p^{k-1}.$$

Example: For $p = 3$ and $k = 2$ ($p^k = 9$):

$$\phi(9) = 9 - 3 = 6.$$

The integers 1, 2, 4, 5, 7, 8 are relatively prime to 9.

3. Formula for $\phi(n)$

Statement: If n has the prime factorization:

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k},$$

then:

$$\phi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

Proof: 1. Start with all integers from 1 to n (n integers total). 2. Exclude integers divisible by each prime factor p_i . The proportion of such integers is $\frac{1}{p_i}$, so the remaining proportion is:

$$1 - \frac{1}{p_i}.$$

3. For all prime factors, adjust for overlaps using the inclusion-exclusion principle. The total proportion of integers that are relatively prime to n is:

$$\prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

4. Multiplying this proportion by n , we get:

$$\phi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

Example: Compute $\phi(12)$:

$$n = 12 = 2^2 \cdot 3^1, \quad \phi(12) = 12 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right).$$

$$\phi(12) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4.$$

4. Proof: $\phi(n)$ is a Multiplicative Function

Statement: If $\gcd(m, n) = 1$, then:

$$\phi(mn) = \phi(m) \cdot \phi(n).$$

Proof: 1. By definition, $\phi(mn)$ counts integers $1 \leq k \leq mn$ such that:

$$\gcd(k, mn) = 1.$$

2. Since $\gcd(m, n) = 1$, the conditions $\gcd(k, m) = 1$ and $\gcd(k, n) = 1$ are independent. 3. Consider the set $S_{mn} = \{k \mid 1 \leq k \leq mn\}$. Divide S_{mn} into m blocks of size n :

$$\{1, 2, \dots, n\}, \{n+1, n+2, \dots, 2n\}, \dots, \{(m-1)n+1, \dots, mn\}.$$

4. In each block, exactly $\phi(n)$ integers satisfy $\gcd(k, n) = 1$. Since there are $\phi(m)$ such blocks, the total number of integers satisfying $\gcd(k, mn) = 1$ is:

$$\phi(mn) = \phi(m) \cdot \phi(n).$$

Example: Compute $\phi(30)$:

$$30 = 2 \cdot 3 \cdot 5, \quad \phi(30) = \phi(2) \cdot \phi(3) \cdot \phi(5).$$

$$\phi(30) = 1 \cdot 2 \cdot 4 = 8.$$

Exercises

Exercise 1: Prove $\phi(n)$ is Always Even for $n > 2$

Proof:

1. From the formula:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

where p are distinct prime factors of n .

2. For $n > 2$, at least one odd prime $p > 2$ divides n , making $n \times \left(1 - \frac{1}{p}\right)$ even.

3. Multiplying this factor with n , $\phi(n)$ is always even for $n > 2$.

Exercise 2: Prove $a \mid b \implies \phi(a) \mid \phi(b)$

Proof:

1. $b = a \cdot k$ as $a \mid b$.

2. Multiplicative property: $\phi(b) = \phi(a \cdot k) = \phi(a) \cdot \phi(k)$

3. Since $\phi(k)$ is an integer, $\phi(a) \mid \phi(b)$.

1 Class 5: Stars and Bars

Bars and Stars Theorem

The **Bars and Stars Theorem** is a combinatorial method to solve the problem of distributing n identical items into k distinct groups.

Theorem Statement

The number of ways to distribute n identical items into k distinct groups is given by:

$$\binom{n+k-1}{k-1}$$

Proof of the Theorem

Consider n identical items and $k-1$ dividers (bars). The total number of objects is $n+k-1$. Arranging these objects in a line gives:

$$\binom{n+k-1}{k-1}$$

ways, as we need to select $k-1$ positions for the dividers among the $n+k-1$ total positions.

subsection*Explanation with String Analogy The problem involves dividing n stars (S) into k groups using $k-1$ bars (B).

1. Arrange n S 's and $k - 1$ B 's in a single string.
2. For example, if $n = 5$ and $k = 3$, one possible string is $SSSSSBB$.
3. The total number of unique permutations of this string is:

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}.$$

Recall

The number of ways to distribute n identical items into k distinct groups is:

$$\binom{n+k-1}{k-1}.$$

Robot Grid Problem

To find the number of unique paths in an $m \times n$ grid, where the robot starts at the top-left and can only move right or down, we use the following approach:

Solution

The robot must make $m - 1$ downward moves and $n - 1$ rightward moves, for a total of:

$$m + n - 2 \text{ moves.}$$

The total number of unique paths is:

$$\binom{m+n-2}{m-1} = \binom{m+n-2}{n-1}.$$

Proof

Each unique path corresponds to an arrangement of $m - 1$ downward (D) and $n - 1$ rightward (R) moves. The number of such arrangements is the number of combinations of $m - 1$ D 's from the total $m + n - 2$ moves:

$$\binom{m+n-2}{m-1}.$$

Explanation with String Analogy

The robot must move $m - 1$ steps downward (D) and $n - 1$ steps to the right (R) to reach the bottom-right corner of an $m \times n$ grid.

1. Arrange $m - 1$ D 's and $n - 1$ R 's in a single string.
2. For example, if $m = 3$ and $n = 4$, one possible string is $DDRRRR$.
3. The total number of unique permutations of this string is:

$$\binom{m+n-2}{m-1} = \binom{m+n-2}{n-1}.$$

Answer

The number of unique paths is:

$$\binom{m+n-2}{m-1}.$$

Proof of Divisibility of $n(n+1)(2n+1)$ by 6

The goal is to prove that $n(n+1)(2n+1)$ is divisible by 6 for all positive integers n .

Step 1: Divisibility by 2

The product $n(n+1)$ is the product of two consecutive integers. One of these integers is always even, so $n(n+1)$ is divisible by 2.

Step 2: Divisibility by 3 (Using Modular Arithmetic)

Consider n modulo 3:

- If $n \equiv 0 \pmod{3}$, then n is divisible by 3, so $n(n+1)(2n+1) \equiv 0 \pmod{3}$.
- If $n \equiv 1 \pmod{3}$, then $n+1 \equiv 2 \pmod{3}$ and $2n+1 \equiv 2(1)+1 = 3 \equiv 0 \pmod{3}$. Hence, $2n+1$ is divisible by 3.
- If $n \equiv 2 \pmod{3}$, then $n+1 \equiv 0 \pmod{3}$, so $n+1$ is divisible by 3.

In all cases, at least one factor among n , $n+1$, or $2n+1$ is divisible by 3.

Step 3: Combined Divisibility

Since $n(n+1)(2n+1)$ is divisible by both 2 and 3, it is divisible by $\text{lcm}(2, 3) = 6$.